A Technical and Mathematical Audit of the RiemannOS v1.1 Framework

Independent Mathematical Audit Team

May 2026

To: Mr. Abdulla Al-Jabri
Subject: Request for Clarifications and Mathematical Derivations

---

1 Introduction

This memo aims to evaluate the mathematical and physical verifiability of the claims presented in the RiemannOS v1.1 framework. It does not assume the framework is correct or incorrect. Instead, it identifies the specific equations and derivations required for these claims to be independently checked and reproduced.

2 Scope and Limitations

The findings in this audit are based strictly on the formulas, definitions, and text excerpts available in the provided v1.1 document.

- The numerical calculations represent partial sums (e.g., using the first 5 Riemann zeros) and do not replace the rigorous verification of the full infinite series.
- All computations are provided for reproducibility, but the audit remains open to revision if the original author provides missing equations, scaling factors, or alternative definitions.
- The goal is to separate numerical coincidences from derived physics, not to dismiss the underlying conceptual ideas.

3 Audit Findings and Required Derivations

3.1 The Z Kernel and Dark Energy (Λ)

Observation: Using the published formula:

$$Z(x) = -\sum_{k=1}^{\infty} \frac{2x}{x^2 + \gamma_k^2}$$

The numerical evaluation of the second derivative at the fifth zero ($x = 32.935062$) using the first 5 zeros yields:

$$Z''_{\text{partial}}(32.935062) \approx 5.504 \times 10^{-5}$$

The text claims a value in the order of 10^{-123}.

Required from Author:

- Provide the complete, exact mathematical definition of $Z(x)$.
- Clarify if there are missing scaling factors, exponential decays, or transformations not shown in the v1.1 text.
- Provide the explicit derivation of the relation $\Lambda = F(Z''(t_5), \gamma_1, \dots)$, including all constants and physical units.
- Explain the discrepancy between the calculated partial value and the claimed order of magnitude.

3.2 Stability and Constants A, B

Observation: The text uses $A = e^{2\pi/\gamma_1}$ and $B = e^{\pi/\gamma_1}$. However, substituting $\gamma_1 = 14.134725$ yields $A \approx 1.56$ and $B \approx 1.25$, not the claimed 10^{15} and 10^{10}. Furthermore, the inequality $V''(Z_*) > 0$ is a local stability condition, not an equation that uniquely fixes A and B.

Required from Author:

- Provide the explicit potential function $V(Z)$.
- Define the equilibrium point Z_*.
- Define the stability operator and the eigenvalue problem if A and B are indeed its eigenvalues.
- Show the rigorous derivation linking the stability condition to the specific numerical values of A and B.

3.3 The Fine Structure Constant (α^{-1})

Observation: The text claims $\alpha^{-1} \approx 137.036$ is derived from γ_7 and γ_9. However, no explicit equation linking these zeros to the constant is provided in v1.1.

Required from Author:

- Provide the exact formula linking Riemann zeros to the fine structure constant.
- Specify all parameters and choices used.
- Clarify whether this formula was derived *a priori* or fitted *after* observing the value 137.036.
- Provide an independent test (e.g., applying the formula to another constant) to prove it is not just numerology.

3.4 Coefficient c_2 and the 2.31 TeV Prediction

Observation: The appendix defines:

$$c_2 = \frac{1}{2} \sum_{k=1}^{\infty} \frac{\ln(\gamma_k)}{\gamma_k^4}$$

Using the first 5 zeros, we calculate $c_2 \approx 4.856 \times 10^{-5}$. This is a dimensionless quantity. The text claims this relates to a 2.31 TeV mass (a 1.2% shift in G_{eff}).

Required from Author:

- Provide the specific energy scale (Scaling Factor) used to convert the dimensionless c_2 into TeV.

- Justify this scale from an independent physical principle, not by fitting it to the target value (2.31 TeV).
- Provide the derivation of the 1.2% shift in G_{eff}.

4 Reproducibility and Verification Requirements

To allow independent review, please provide:

1. The complete intermediate derivations connecting the Z kernel to the physical observables.
2. A detailed explanation of the mathematical conditions required for the convergence of the series and the validity of the term-by-term differentiation.
3. The source code (or detailed algorithms) used to generate the large-scale numerical results, including precision and software versions.
4. At least one prediction that can be tested without re-calibrating the parameters to the data it aims to explain.

5 Conclusion

This memo does not seek to reject RiemannOS, but to establish the boundary between mathematical hypothesis, derived result, and physical prediction. Providing the requested equations and independent tests will enable a definitive scientific evaluation and could pave the way for a verifiable v2.0.

Guiding Principle:
The Idea → Mathematical Derivation → Numerical Result → Physical Interpretation → Independent Test.